



Giving-Hand

Presented By

**Morris Gergis
Joustina Ashraf
Omar Khaled
Omar Saad
Mariam Ahmed
Zeyad Hussein**

Supervised Under

Marwa Sabry



Table of Content

Introduction
Context, DFD's, ERD, Uml, Use Case
Problem gap
Objectives
Project scope
Competitive advantages
Live demo



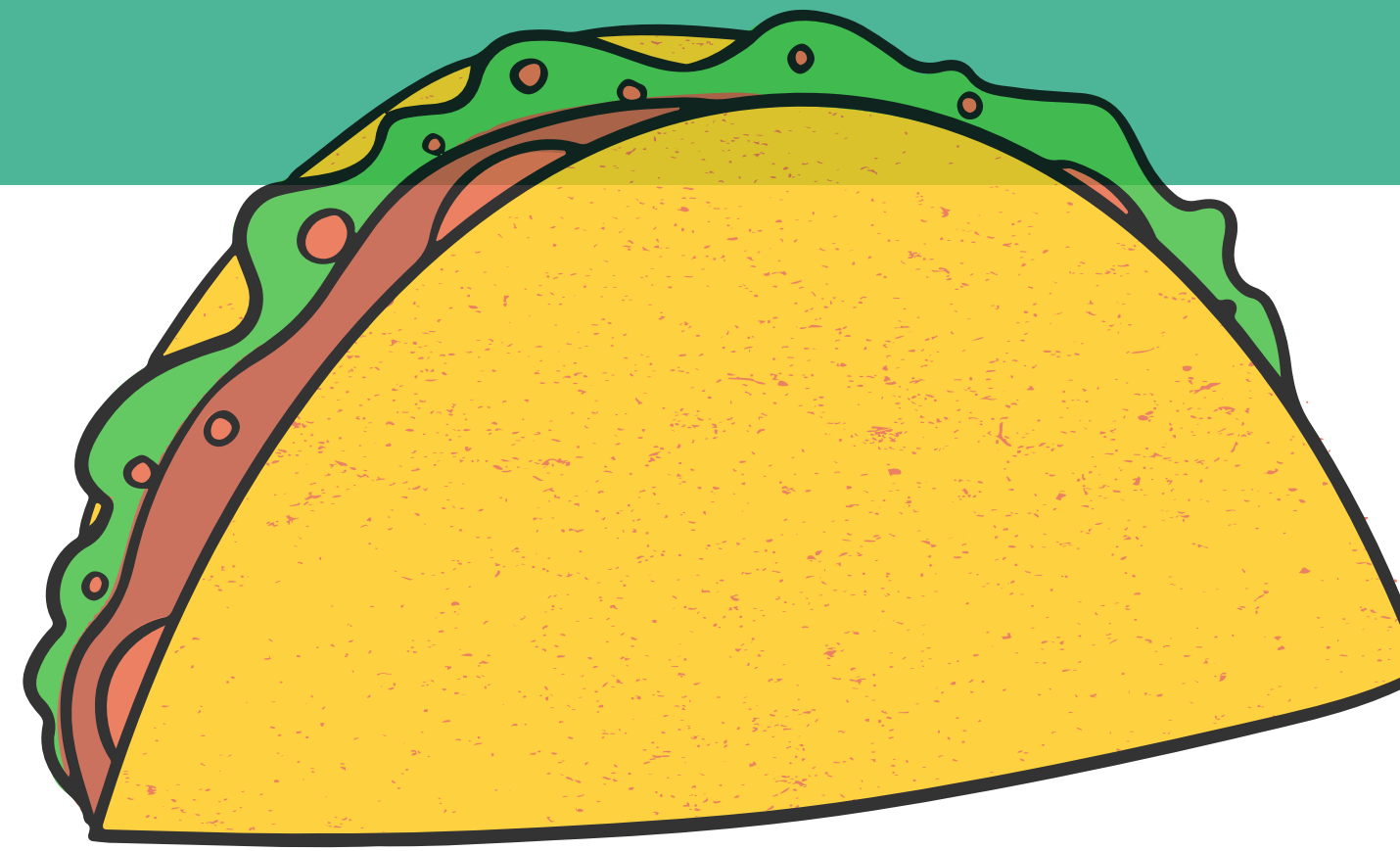


Giving Hand is a mobile and web platform created to **reduce food waste**. It connects places that have extra food—like **hotels, restaurants, supermarkets, and factories**—with **charities** and **shelters** that needs it. This helps make sure good food doesn't go to waste and instead reaches people who needs it.

About Us



Problem Gap



- **Edible food waste**

- **Expired food adds to landfills**

- **Charities face ongoing food shortages**

Objectives



Support charities & shelters



Turn expired food into fertilizer



Promote CSR



Timely food donations



Scalable across Egypt



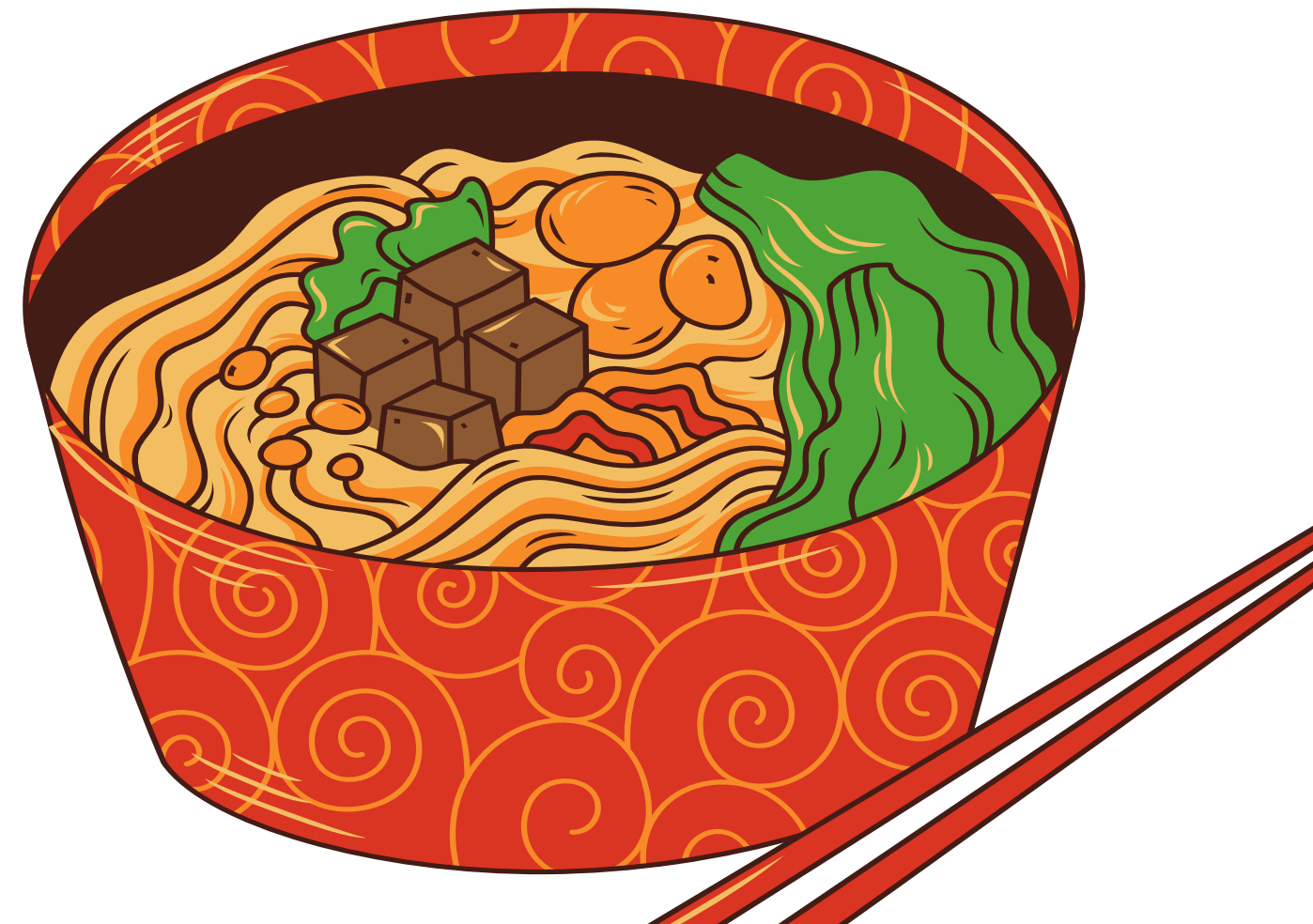
Builds trust and transparency

Competitive advantage

Support charities & shelters

Promote CSR

Scalable across regions

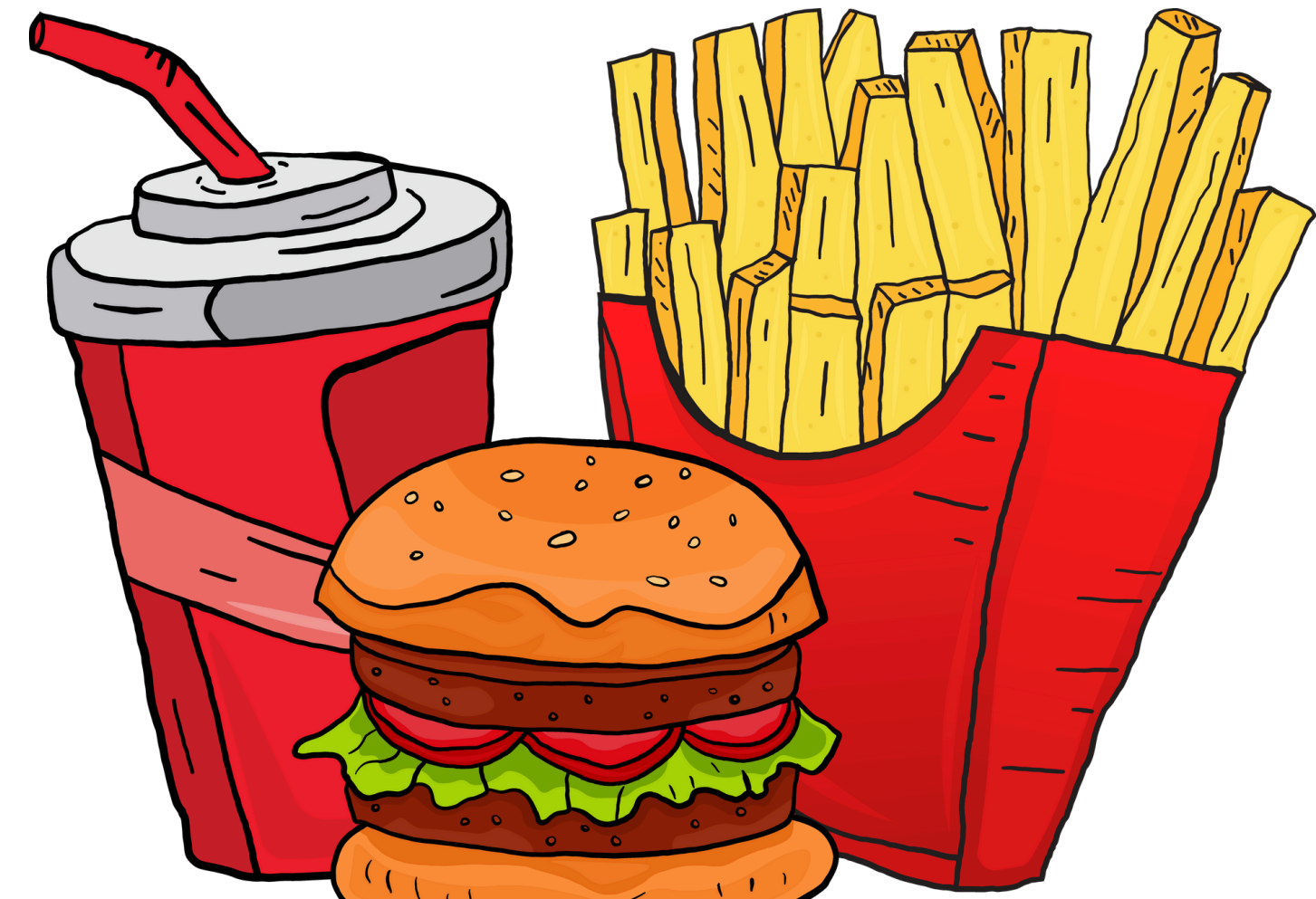


Competitive advantage

Turn expired food into fertilizer

Timely food donations

Builds trust and transparency



Project scope



Inclusions:

- web app
- *Connects* food donors (hotels, restaurants, supermarkets, factories) with charities & shelters
- Real-time donation system and tracking



Exclusions:

- No food delivery by us
- No logistics
- No food handling or preparations



Entities

Hotels



Supermarkets



Factories



Restaurants



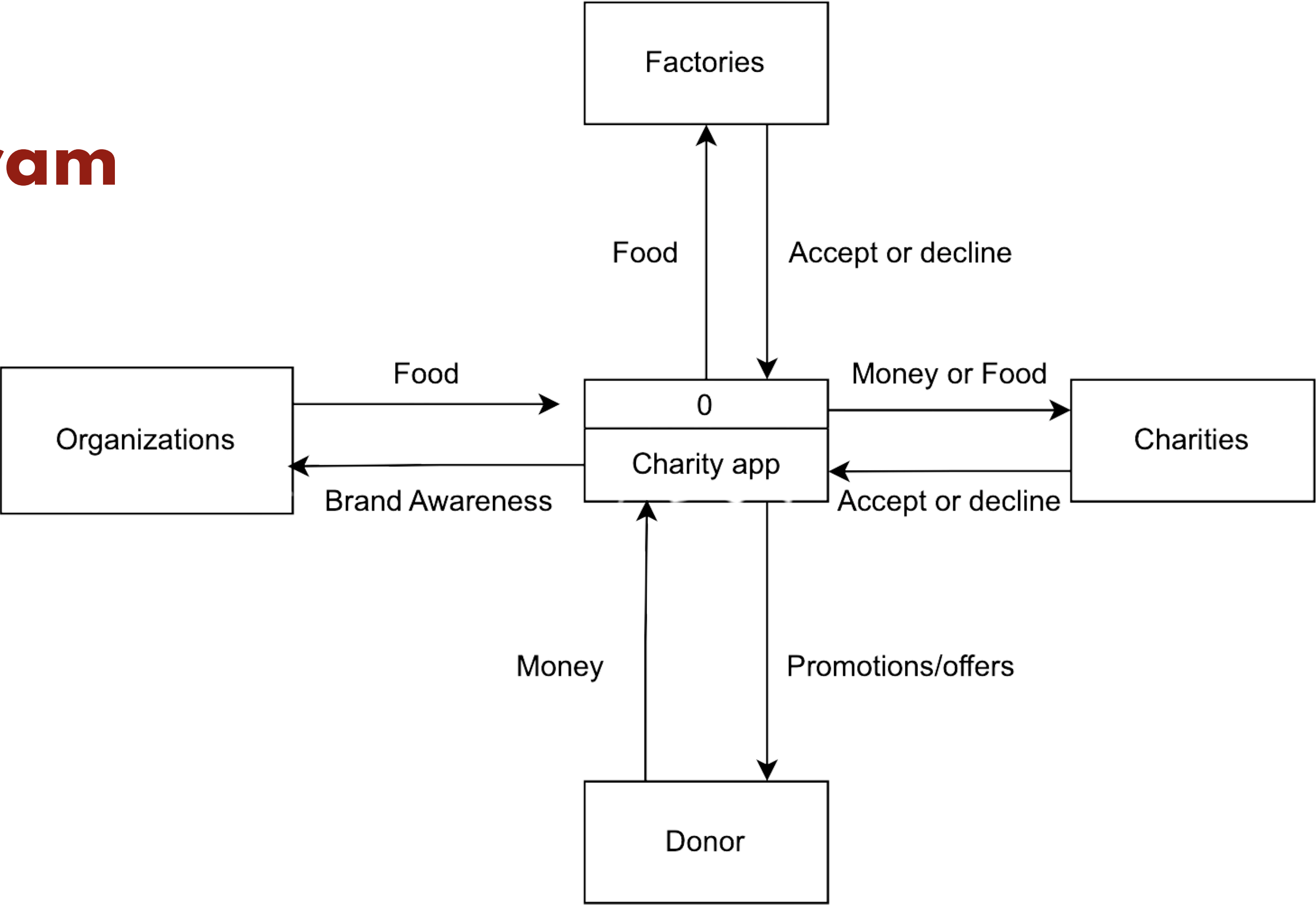
Shelters



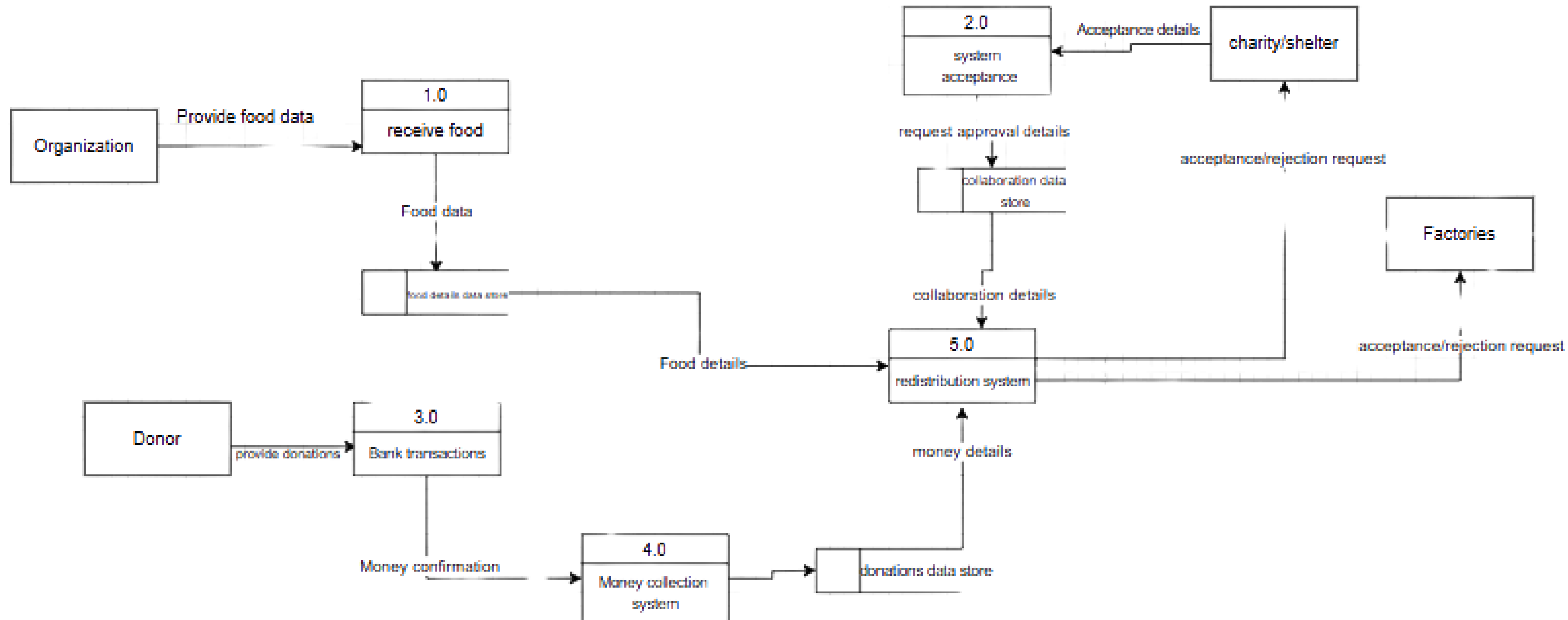
Charities



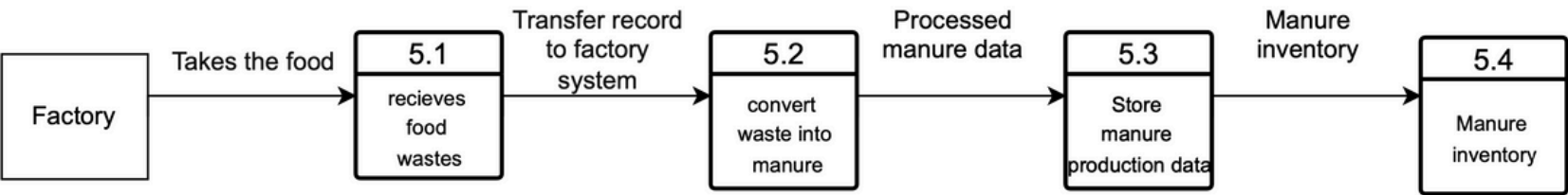
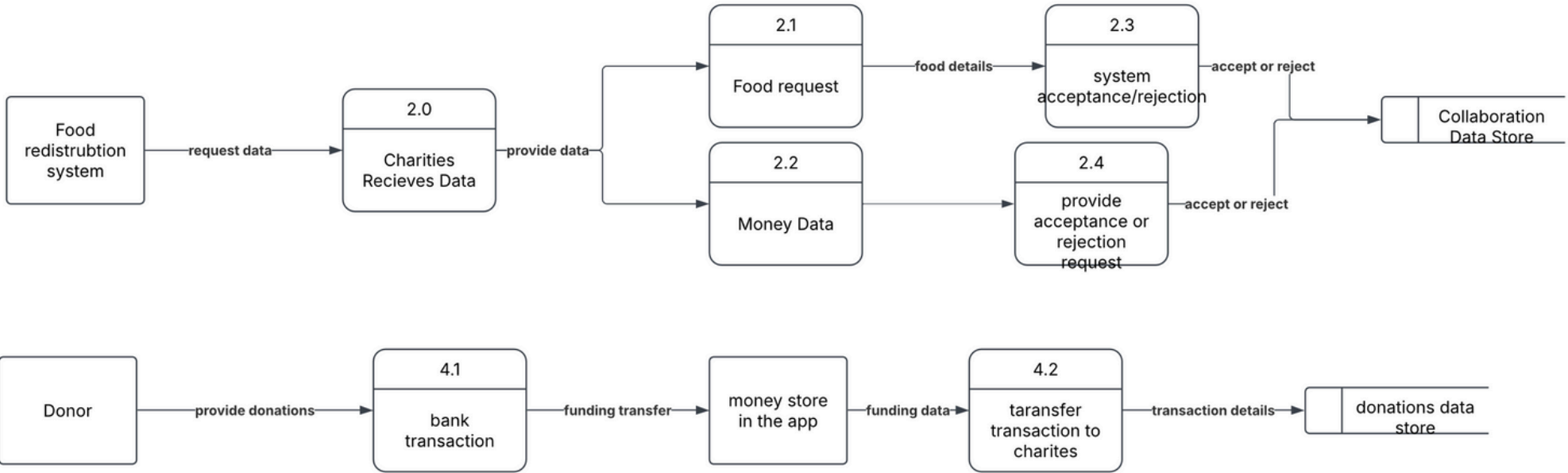
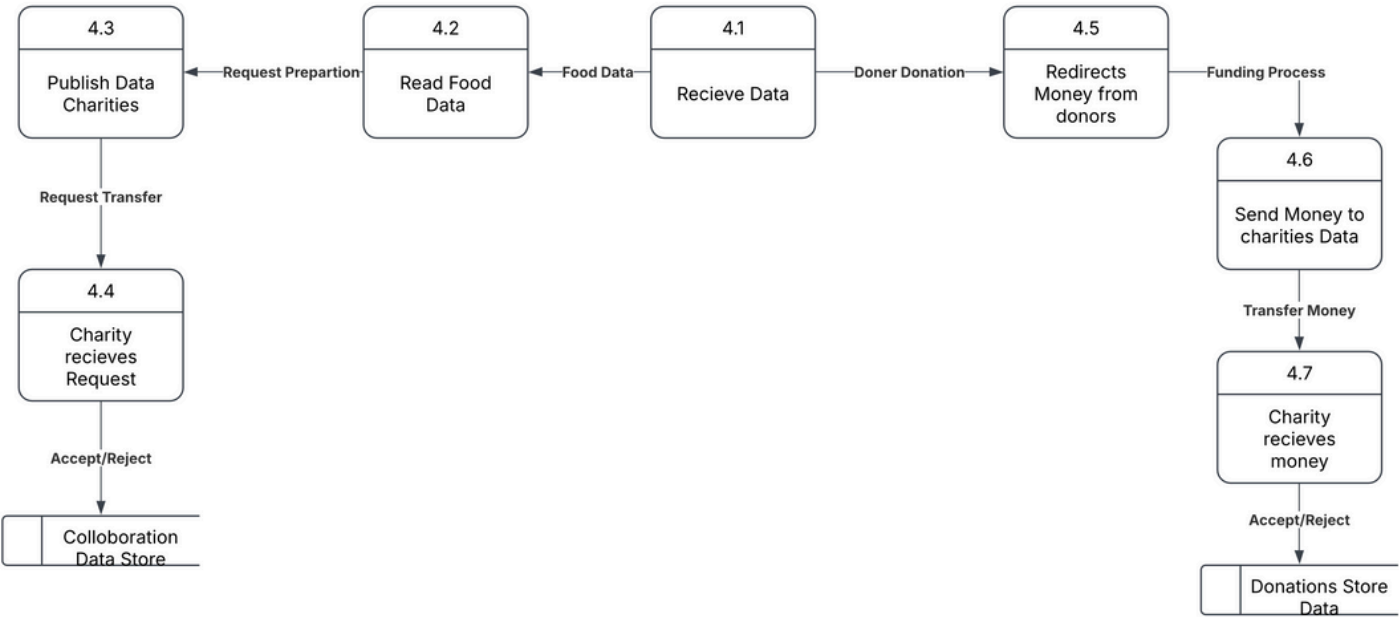
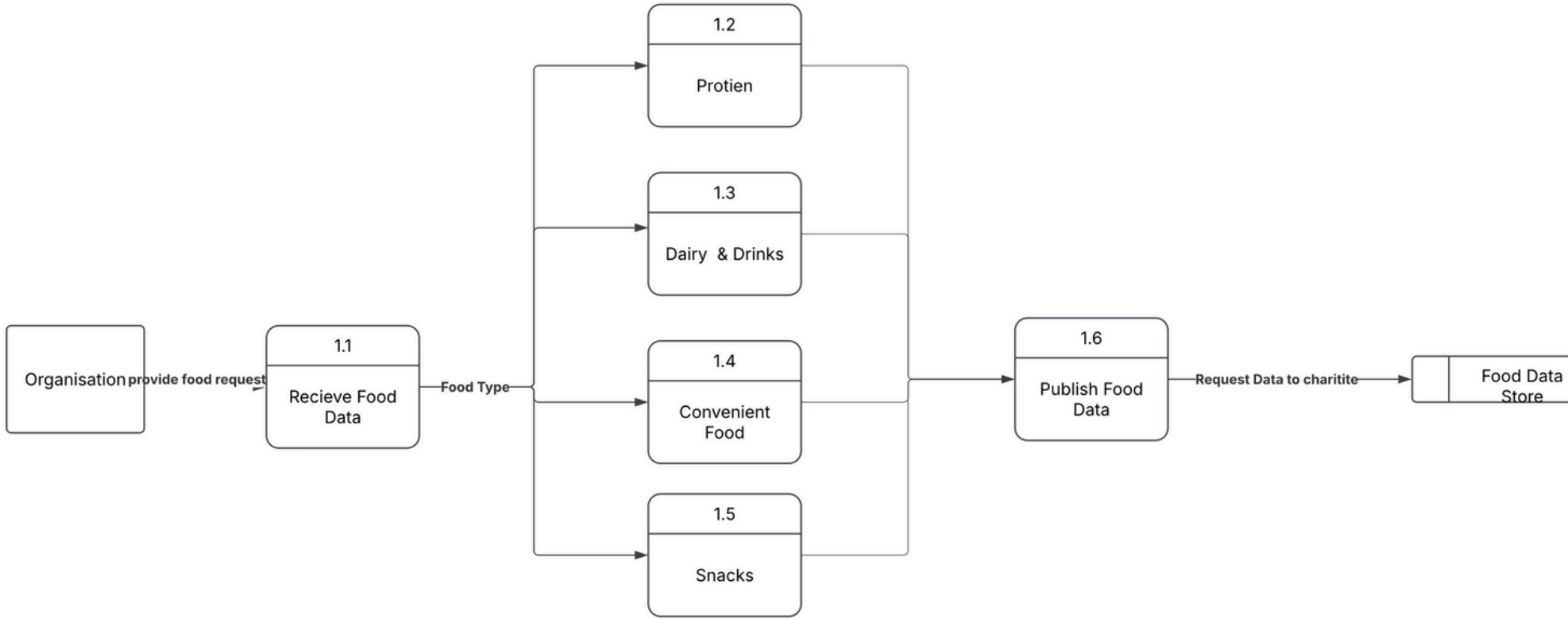
Context diagram



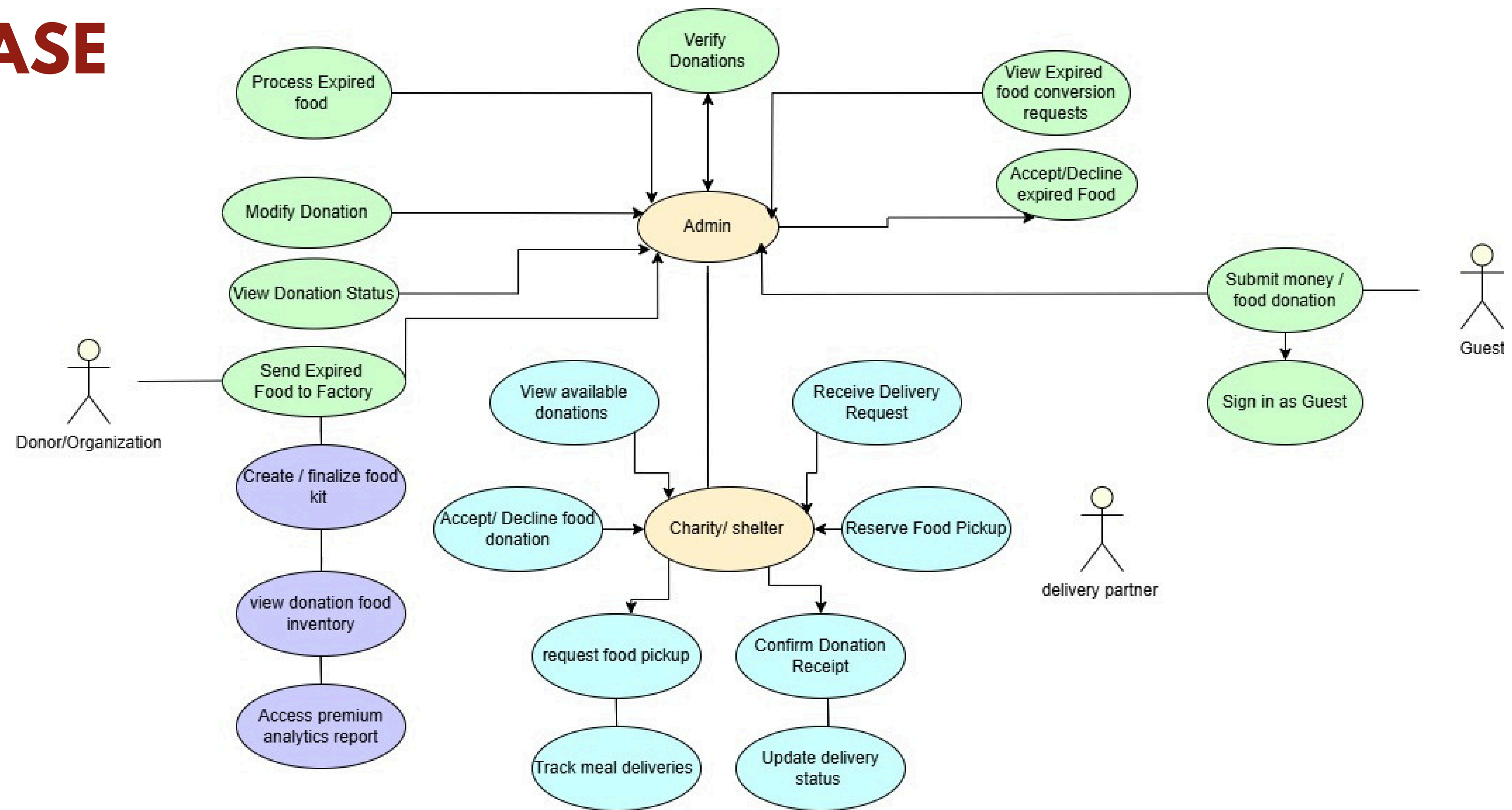
DFD 0



DFD 1



USE CASE



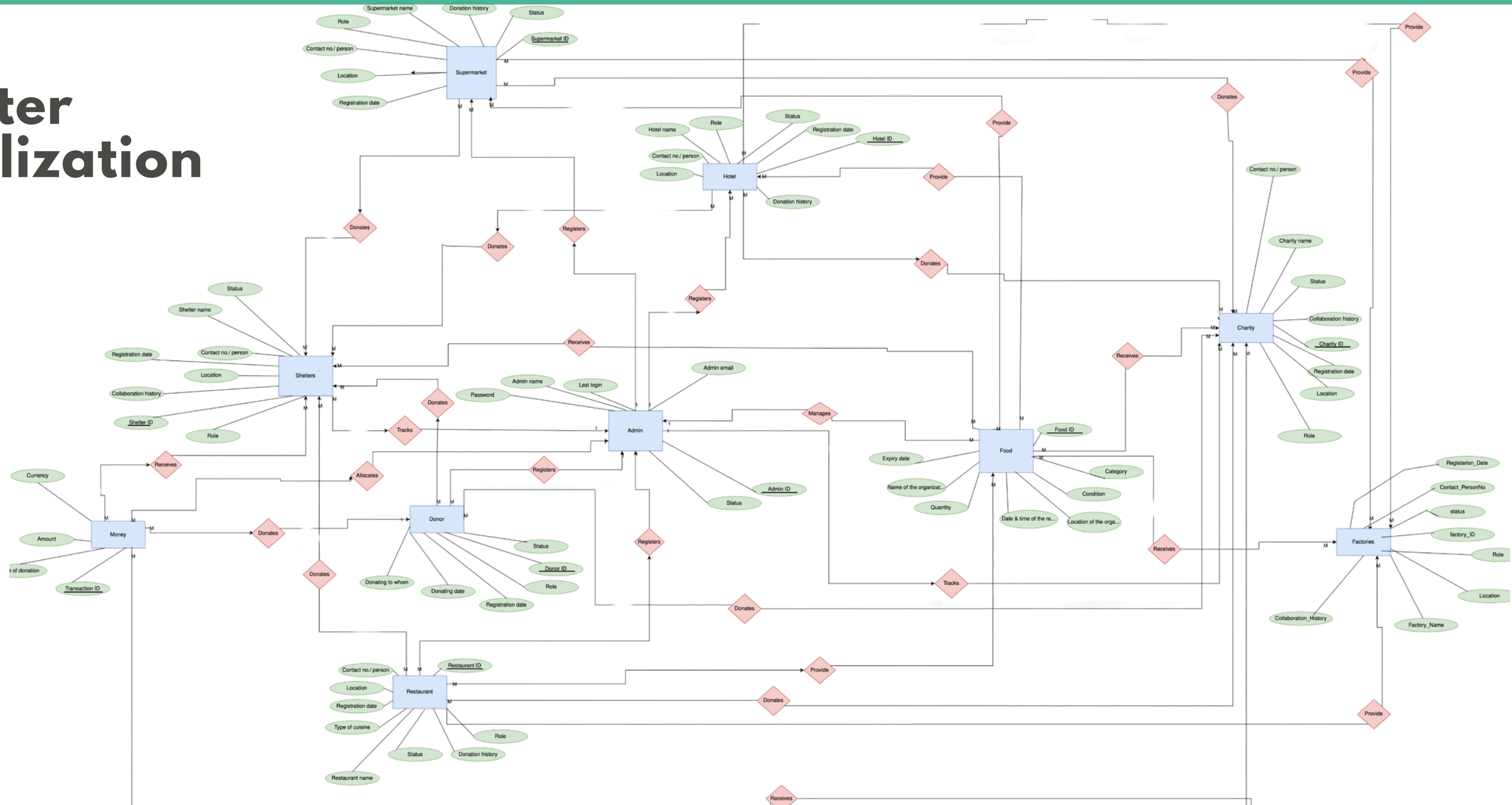
Data Dictionary

Payments						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	Payment ID
donation_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Linked Donation
method	ENUM('visa','fawry')	None	Yes	N/A	Value must be one of the allowed methods	Payment method
status	ENUM('pending','completed','failed')	None	Yes	N/A	Value must be one of the allowed statuses	Payment status
transaction_details	TEXT	None	No	65535	Text up to 65535 characters	Details or reference
Discounts						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	Discount ID
user_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	User receiving discount
org_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Organization offering discount
percentage	DECIMAL(5,2)	None	Yes	5,2	Numeric with 5 digits and 2 decimal places	Discount percent
discount_code	VARCHAR(50)	None	Yes	50	Max length 50	Generated code
Notifications						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	Notification ID
ticket_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Related ticket
recipient_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Notified recipient
status	ENUM('sent','viewed','unread')	None	Yes	N/A	Value must be one of the allowed statuses	Notification status
History						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	History ID
recipient_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Related recipient
collab_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Linked collaboration
recorded_at	TIMESTAMP	None	Yes	N/A	Must be a valid timestamp	Time of record
Reports						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	Report ID
org_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Organization ID
data_summary	TEXT	None	Yes	65535	Text up to 65535 characters	Summary or analytics output
created_at	TIMESTAMP	None	Yes	N/A	Must be a valid timestamp	Generated on
Donations						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	Donation ID
user_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Donor's user ID
recipient_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Charity receiving donation
amount	DECIMAL(10,2)	None	Yes	10,2	Numeric with 10 digits and 2 decimal places	Donation amount

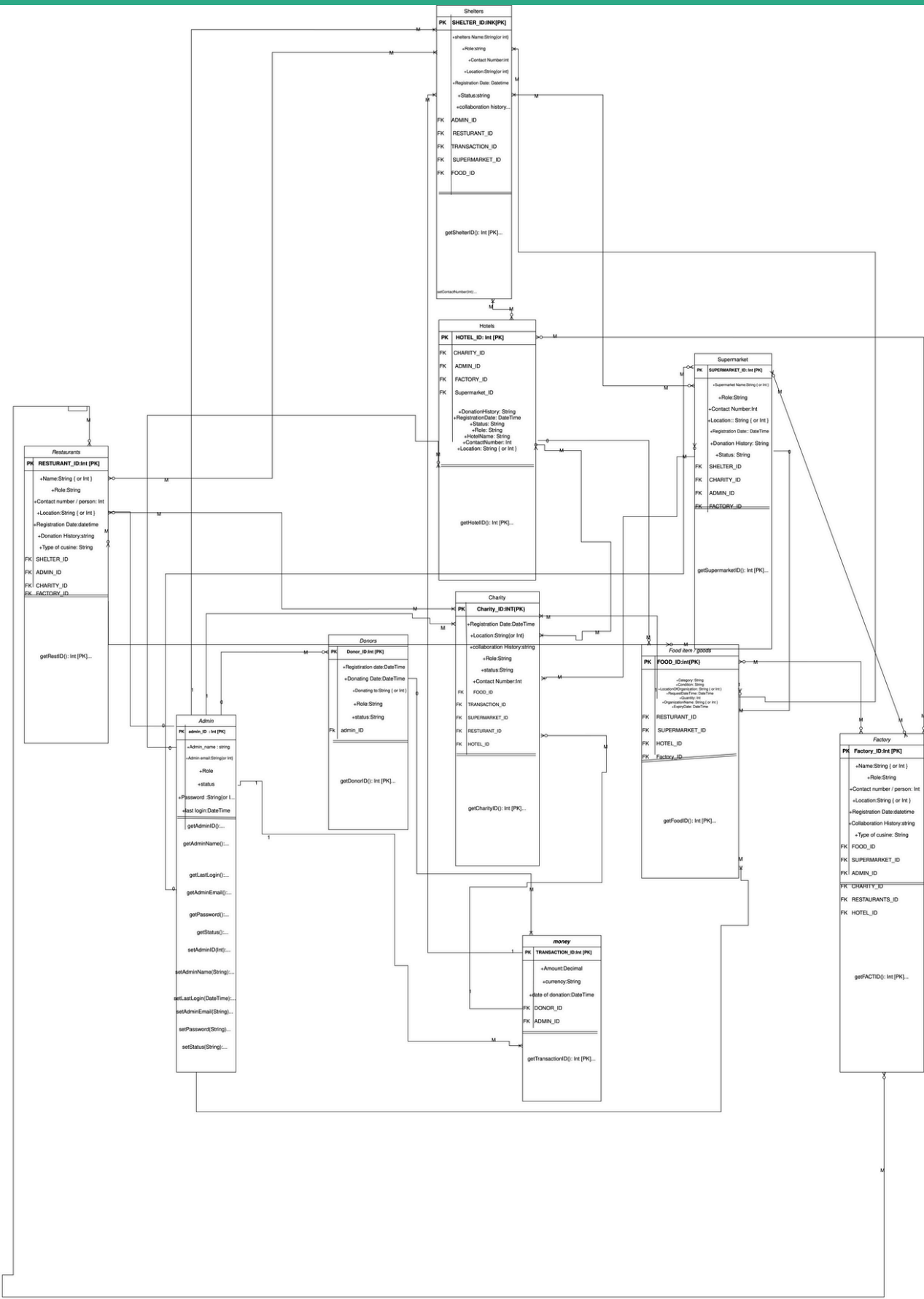
Data Dictionary

Users						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	Unique user identifier
username	VARCHAR(50)	Unique	Yes	50	Max length 50	User's login name
email	VARCHAR(100)	Unique	Yes	100	Max length 100	User's email address
password_hash	VARCHAR(255)	None	Yes	255	Max length 255	Hashed password
role	ENUM('organization', 'recipient')	None	Yes	N/A	Value must be one of 'organization', 'recipient'	User's role
created_at	TIMESTAMP	None	Yes	N/A	Must be a valid date/time	Account creation time
Organizations						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	Organization ID
user_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Link to Users table
name	VARCHAR(100)	None	Yes	100	Max length 100	Organization name
type	ENUM('hotel', 'restaurant', 'shelter')	None	Yes	N/A	Value must be one of 'hotel', 'restaurant', 'shelter'	Type of organization
location	VARCHAR(255)	None	Yes	255	Max length 255	Address or area
phone	VARCHAR(20)	None	No	20	Max length 20	Contact phone
Recipients						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	Recipient ID
user_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Link to Users table
name	VARCHAR(100)	None	Yes	100	Max length 100	Charity or shelter name
category	ENUM('children', 'elderly', 'disabled')	None	Yes	N/A	Value must be one of 'children', 'elderly', 'disabled'	Charity focus area
Tickets						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	Ticket ID
org_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Link to Organizations table
food_type	VARCHAR(100)	None	Yes	100	Max length 100	Type of food
category	VARCHAR(100)	None	Yes	100	Max length 100	Food category
weight	DECIMAL(10,2)	None	Yes	10,2	Numeric with 10,2 precision	Weight in kg
expiry_date	DATE	None	Yes	N/A	Must be a valid date/time	Food expiry
notes	TEXT	None	No	65535	Text up to 65535 characters	Additional info
status	ENUM('pending', 'accepted', 'rejected')	None	Yes	N/A	Value must be one of 'pending', 'accepted', 'rejected'	Ticket status
created_at	TIMESTAMP	None	Yes	N/A	Must be a valid date/time	Ticket creation time
Collaborations						
Field name	Type	Key	Required	Size	Validation	Description
id	UUID	Primary Key	Yes	36	Must be a valid UUID	Collaboration ID
ticket_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Linked Ticket
recipient_id	UUID	Foreign Key	Yes	36	Must be a valid UUID	Recipient who accepted
delivery_method	ENUM('pickup', 'organization_delivery')	None	Yes	N/A	Value must be one of 'pickup', 'organization_delivery'	Method of delivery
collaboration_date	DATE	None	Yes	N/A	Must be a valid date/time	Date of collaboration

ERD after normalization



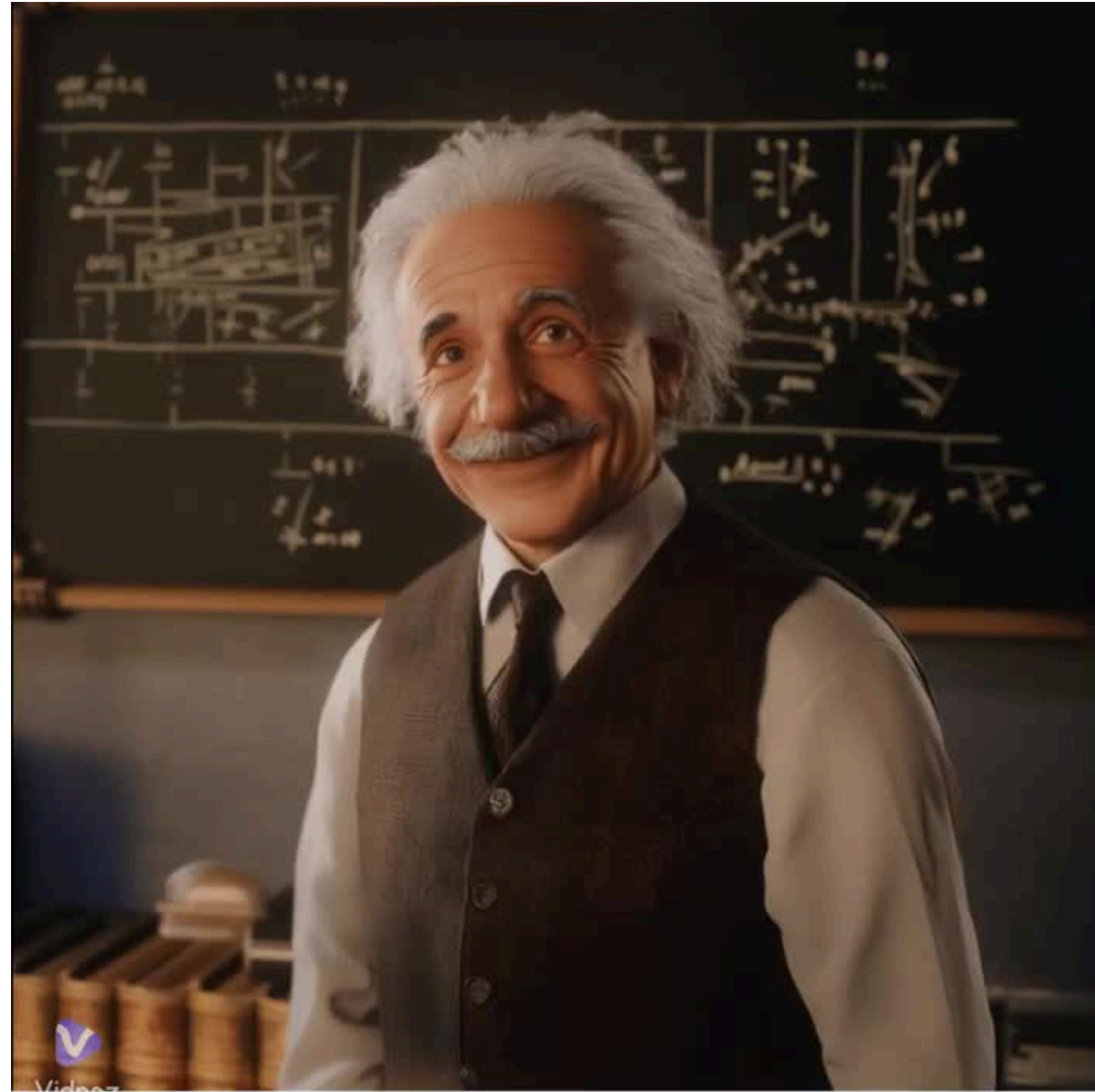
UML



Live Demo



Reliable source



Thank you



Special thanks for

Dr. Moustafa Kamel

Dr. Zeinab Nawar

Dr. Marwa Sabry

Dr. Alshimaa

Dr. Anas

for their constant support